1. EQUIPMENT IDENTITY AND PURPOSE

Machine ID: M-HP200
Model: HP-200
Description: Hydraulic forming press.
The HP-200 is a hydraulic forming press engineered for precision component forming under high hydraulic pressure.
Operating fluid: H46 hydraulic oil.

2. OPERATING REFERENCE VALUES

- Normal hydraulic pressure: 120 to 140 bar during the forming stage.
- Ready oil temperature: 35°C to 55°C (required temperature range for readiness).
- High-temperature warning threshold: Above 60°C.
- High-temperature stop threshold: 65°C or higher (immediate cycle cutoff).
- Reservoir oil level: Must remain between marked MIN and MAX level indicators.
- Low-pressure trigger threshold: Below 110 bar for more than 5 seconds during forming stage.

3. MACHINE OPERATING STATES

- OFF: Machine is completely powered down.
- READY: Starting preconditions are met (guard-closed indicator active, reservoir oil level between MIN and MAX, oil temperature between 35°C and 55°C). Note: A normal pressure reading in READY does not prove that pressure will remain normal during FORMING.
- FORMING: Machine actively delivers hydraulic tonnage to form parts.
- HOLD: Cycle paused or inhibited due to fault condition, awaiting review.

4. DIAGNOSTIC PRINCIPLES

A pressure reading alone cannot identify which individual component has failed. Low pressure does not automatically indicate that the hydraulic pump is damaged or worn.

1. OPERATOR PRE-CYCLE REVIEW

Prior to initiating a forming cycle, the operator must verify:

- Machine ID display and machine state indication (must show READY).
- Guard-closed safety indicator. Never bypass safety interlocks under any circumstances.
- Reservoir sight-glass oil level (between MIN and MAX).
- Oil temperature indication (must be in ready range: 35°C to 55°C).

If any indication is abnormal, hold the cycle and report immediately.

2. ALARM CODES REFERENCE

- E17: Low hydraulic pressure. Triggered when pressure drops below 110 bar for more than 5 seconds during FORMING. Normal forming pressure is 120-140 bar. Required response: Record hydraulic pressure and current cycle stage; place machine on HOLD; request inspection checks. Note: E17 does not prove the pump has failed.
- E24: High oil temperature warning. Warning triggered above 60°C; automatic stop occurs at 65°C or above. (For example, at 62°C, temperature is in warning range: machine must be placed on HOLD and cooling-system inspection requested).
- E31: Guard-open indication. Safety interlock triggered. Cycle inhibited. Never bypass interlocks.
- E42: Return-filter restriction. Restriction detected in hydraulic return filter (RF-20). Operator must record indicator status and request filter inspection.

3. TEMPERATURE BEHAVIOR RULES

- 35°C to 55°C: Within normal ready range.
- Above 60°C: Warning condition.
- 65°C or higher: High-temperature stop.

Note: The manual does not specify a guaranteed cooldown duration.

1. FAULT REPORTING REQUIREMENTS

When reporting an HP-200 operational fault or alarm, record:

- Machine ID (M-HP200), date, and exact timestamp.
- Active alarm code (e.g., E17, E24, E31, E42).
- Displayed hydraulic pressure and oil temperature.
- Active cycle stage and reservoir oil level.
- Visible external leakage observations.
- Whether the condition has previously occurred.

2. REPEATED FAULT ESCALATION

If low hydraulic pressure (E17) occurs repeatedly after documented checks have been reviewed:

- Maintain machine on HOLD.
- Escalate issue to authorized maintenance, referencing prior incident records.
- Warning: Never adjust or increase the hydraulic pressure setting just to suppress an alarm.

3. ROLES AND RESPONSIBILITIES

- Operator: Observes and records symptoms, active alarms, and operating parameters.
- Authorized Maintenance: Evaluates technical symptoms and conducts physical inspections.
- Designated Supervisor: Reviews maintenance findings and records the official return-to-service decision.

4. EXCLUDED INFORMATION / GUIDANCE LIMITS

The HP-200 manuals do not provide:

- Hydraulic pump replacement prices or spare parts costs.
- Component supplier contacts.
- Fastener tightening torques.
- Electrical wiring diagrams or internal hydraulic schematics.
- Internal component rebuild procedures.
- Pressure regulator modification steps.
- Guaranteed cycle times or service contract coverage.